DSA ABSTRACT

Resume Search and Candidate Screening System

Companies receive a large number of resumes for job openings, making manual screening time-consuming and difficult. The proposed Resume Search and Candidate Screening System provides an automated approach for searching and ranking candidates according to recruiter-defined requirements. The system maintains a structured corpus of resume records containing candidate details, education, experience, skills and a short professional summary. During preprocessing, skill names are normalized to a consistent format. An inverted index implemented using a HashMap is created to associate each skill with the candidates possessing it, allowing efficient retrieval of relevant candidates. The system calculates a relevance score based primarily on the percentage of required skills matched, with an optional experience component, and uses sorting to rank candidates. The final output presents candidates in descending order of relevance. The prototype uses a self-created synthetic corpus of 40 resumes for academic demonstration and can be extended to larger real-world datasets and additional screening criteria.